

Renewable Energy Sources — Unit 1 & 2 (Crux Notes)

UNIT 1: RENEWABLE ENERGY SOURCES

Energy — Core Definition

Energy = ability to do work; **Work** = transfer of energy from one form to another.

Forms: Thermal, Radiant, Mechanical, Electrical, Chemical, Nuclear **Two states:** Potential (stored) & Kinetic (working)

Energy Classification (3 mandate splits)

Basis	Types	Key point
Origin	Primary (coal, oil, gas, nuclear) vs Secondary (electricity, steam)	Primary converted → secondary
Market	Commercial (priced: coal, oil, electricity) vs Non-commercial (firewood, dung — not accounted)	Non-commercial = "traditional fuels"
Depletion	Renewable (wind, solar, geothermal, tidal, hydro) vs Non-renewable (fossil fuels)	Renewable = inexhaustible, low pollution

Ideal energy source criteria: perennial, widely distributed, non-polluting, cost-effective & safe.

India benchmark data: per-capita electricity ≈ **1208–1371 kWh** (India) vs **5985 kWh** (China) vs **~15,000 kWh** (developed nations) vs **~59,629 kWh** (Iceland, highest).

India RE Capacity (Feb 2023 snapshot — mandate figures)

Total RE (incl. large hydro) = 178.79 GW

- Wind: 42.6 GW · Solar: 66.7 GW · Biomass/Co-gen: 10.2 GW · Small Hydro: 4.94 GW · WtE: 0.55 GW · Large Hydro: 46.85 GW

India's targets: carbon intensity ↓ 45% by 2030 · 50% cumulative installed capacity from RE by 2030 · **Net-zero by 2070**

Global Energy Crisis — Causes (mandate list)

1. **Surge in demand** (1970: 204 Quadrillion BTU; +50% projected by 2030)
2. **Tighter supply**
3. **Political uncertainty** in oil-producing regions (→ *Resource Nationalism*)
4. **Lack of diversity of resources**

Fossil-fuel share of world energy: **Oil 37% + Coal 25% + Gas 23% = 85%**

Fossil Fuel Reserve Life (mandate numbers)

Fuel	Estimated life
Oil	~50 years
Natural Gas	~50–60 years
Coal	~130 years

Shortcomings of Conventional Sources

- Depleting non-renewable stock
- Air pollution (NO_x, SO₂ → acid rain), GHG emissions (CO₂ → global warming)
- Nuclear: radioactive waste, high risk (Fukushima 2011, Chernobyl)
- Heavy T&D loss, limited reach, high dependency = national security risk

Alternative Sources — Pros/Cons Snapshot

Source	Pros	Cons
Solar	Inexhaustible, free, zero emission	Inconsistent, inefficient, high capex
Nuclear	Lowest fatality/unit energy, low-carbon (~10% global electricity, 2017)	Expensive, radioactive waste, accident risk, non-renewable
Tar Sands	Alternative liquid fuel source	+15% CO ₂ vs conventional oil; 5.9 gal water/gal fuel
Oil Shale	Usable as direct fuel	Costlier extraction, high environmental cost
Tidal	Highly predictable (vs wind/solar)	High cost, limited suitable sites
Geothermal	Reliable, growing ~5%/yr	Site-restricted (12.8 GW global, 2015)

UNIT 2: SOLAR ENERGY

Solar Constant (mandate value)

$$I_{sc} = 1367 \text{ W/m}^2$$

Defined as flux on a plane $\perp$ to sun's rays at **1 AU**, outside Earth's atmosphere.

Radiation Terminology (mandate definitions)

- **Extraterrestrial radiation:** radiation falling on Earth outside the atmosphere ($\approx$ constant)
- **Terrestrial radiation:** radiation reaching Earth's surface after atmospheric absorption/scattering
- **Direct (beam) radiation:** reaches surface undiffused
- **Diffuse radiation:** scattered/absorbed by atmosphere, clouds, aerosols
- **Global radiation:** Direct + Diffuse

$$I_{global} = I_{beam} + I_{diffuse}$$

Atmospheric interaction (4 mechanisms): Reflection, Scattering, Absorption, Transmission

Solar Radiation Geometry — Key Formulas

1. Solar Time (Local Apparent Time)

$$LAT = IST \pm 4(L_{st} - L_{loc}) + E$$

- L_{st} = standard meridian longitude, L_{loc} = local longitude
- + if standard meridian is **west** of prime meridian; – if **east**
- E = Equation of Time (correction, in minutes)

2. Declination angle (δ) — angular position of sun relative to equatorial plane:

$$\delta = 23.45^\circ \sin \left[\frac{360}{365}(284 + n) \right]$$

where n = day number of year (from Jan 1)

3. Hour Angle (ω) — angular displacement of sun from local meridian:

$$\omega = 15^\circ \times (\text{Solar Time in hours} - 12)$$

- $\omega = 0^\circ$ at solar noon; positive in forenoon, negative in afternoon
- 15° per hour (Earth rotates 360° in 24 h)
- At 06:00 $\rightarrow \omega = +90^\circ$; at 18:00 $\rightarrow \omega = -90^\circ$

4. Sunset Hour Angle & Day Length

$$\omega_s = \cos^{-1}(-\tan \phi \tan \delta)$$

$$t_d = \frac{2}{15} \cos^{-1}(-\tan \phi \tan \delta) \text{ hours}$$

where ϕ = latitude of location. (Positive root → sunrise hour angle, negative root → sunset hour angle)

5. Angle of Incidence (beam radiation on a tilted, arbitrarily-oriented surface) — the master (general) formula:

$$\cos \theta = \sin \delta \sin \phi \cos \beta - \sin \delta \cos \phi \sin \beta \cos \gamma - \cos \delta \cos \phi \cos \beta \cos \omega - \cos \delta \sin \phi \sin \beta \cos \gamma \cos \omega$$

Variable key (all angles in degrees unless noted; result of $\cos \theta$ is dimensionless):

Symbol	Name	Physical meaning	Unit
θ	Angle of incidence	Angle between the sun's beam and the normal to the tilted surface	degree (°)
δ	Declination angle	Angular position of the sun relative to the plane of the equator (positive in N hemisphere)	degree (°)
ϕ	Latitude	Latitude of the location (positive for Northern hemisphere)	degree (°)
β	Slope / Tilt angle	Angle between the collector plane and the horizontal	degree (°)
γ	Surface azimuth angle	Deviation of the projection of the surface normal from local meridian; 0° = due south, positive = west of south, negative = east of south	degree (°)
ω	Hour angle	Angular displacement of the sun east/west of local meridian due to earth's rotation (15° per hour)	degree (°)
n	Day number	Day of the year counted from Jan 1 (used to compute δ)	integer (days)

Special cases (reductions of the master formula — mandate shortcuts):

- Horizontal surface ($\beta = 0^\circ$):

$$\cos \theta_z = \sin \delta \sin \phi + \cos \delta \cos \phi \cos \omega$$

(here θ_z = **zenith angle**)

- Vertical surface ($\beta = 90^\circ$):

$$\cos \theta = -\sin \delta \cos \phi \cos \gamma + \cos \delta \sin \phi \cos \gamma \cos \omega + \cos \delta \sin \gamma \sin \omega$$

- Tilted surface facing due south ($\gamma = 0^\circ$):

$$\cos \theta = \sin \delta \sin(\phi - \beta) + \cos \delta \cos(\phi - \beta) \cos \omega$$

Solar Radiation Spectrum (mandate table)

Type	Wavelength	Main Use
Radio	> 1 mm	Communication
Microwave	1 mm – 1 cm	Radar, GPS
Infrared	700 nm – 1 mm	Heat sensing
Visible	400 – 700 nm	Vision, lighting
UV	10 – 400 nm	Disinfection
X-ray	0.01 – 10 nm	Medical
Gamma	< 0.01 nm	Cancer therapy

Measurement Instruments (mandate terms)

- **Pyranometer** — measures **global** radiation (horizontal/inclined surface); with shading ring → diffuse only
- **Pyrheliometer** — measures **beam** radiation only (narrow collimator tube, normal incidence); sensitivity $\approx 8 \text{ mV/W/m}^2$
- **Sunshine Recorder** — measures **sunshine duration (hours)**; glass sphere $\approx 10 \text{ cm dia.}$

Solar Collector Classification

$$\text{Collector Efficiency} = \frac{\text{Useful energy absorbed and transferred}}{\text{Energy incident on collector}}$$

$$\text{Concentration Ratio (CR)} = \frac{\text{Aperture area}}{\text{Receiver area}}$$

Type	CR	Temp. range
Flat plate	1	< 100 °C
Line focus	up to 100	150–300 °C
Point focus	thousands	500–1000 °C

Non-concentrating: collector area = absorber area; uses beam + diffuse radiation

Concentrating: absorber < interceptor; mainly beam radiation; needs sun-tracking

Flat Plate Collector — Components (mandate list)

1. Absorber plate (Cu/steel/plastic, black coated)
2. Glazing (transparent cover — glass/plastic)
3. Heat-transport fluid (air/water/oil)
4. Insulation (back & sides)
5. Flow passages
6. Enclosure

Evacuated Tube Collector (ETC)

- Double glass tube, **vacuum** between → minimal convective/conductive loss
- Inner tube: selective coating (high absorption, low reflection)

Concentrating Collectors

- **Acceptance angle:** total angle sun can move through without missing receiver (60° acceptance → CR = 2)
- **Compound Parabolic Concentrator (CPC):** non-imaging, CR = 3–7, highest acceptance angle
- **Parabolic Dish:** dual-axis tracking, up to **1000 °C**, often paired with Stirling engine

Other Solar Applications (mandate terms only)

- **Solar Pond:** salinity-gradient water body; bottom layer up to **90 °C**, top ~30 °C
- **Solar Distillation:** desalination via evaporation–condensation in a "solar still"
- **Solar Dryer:** Direct (sun on substance) vs Indirect (heated air passed over substance)
- **Solar Water Heater:** Passive (thermosiphon, tank above collector) vs Active (pump-driven, tank below collector)

Compiled from RES Unit 1 & 2 slide deck — formalities and formulas prioritized per mandate term requirements.